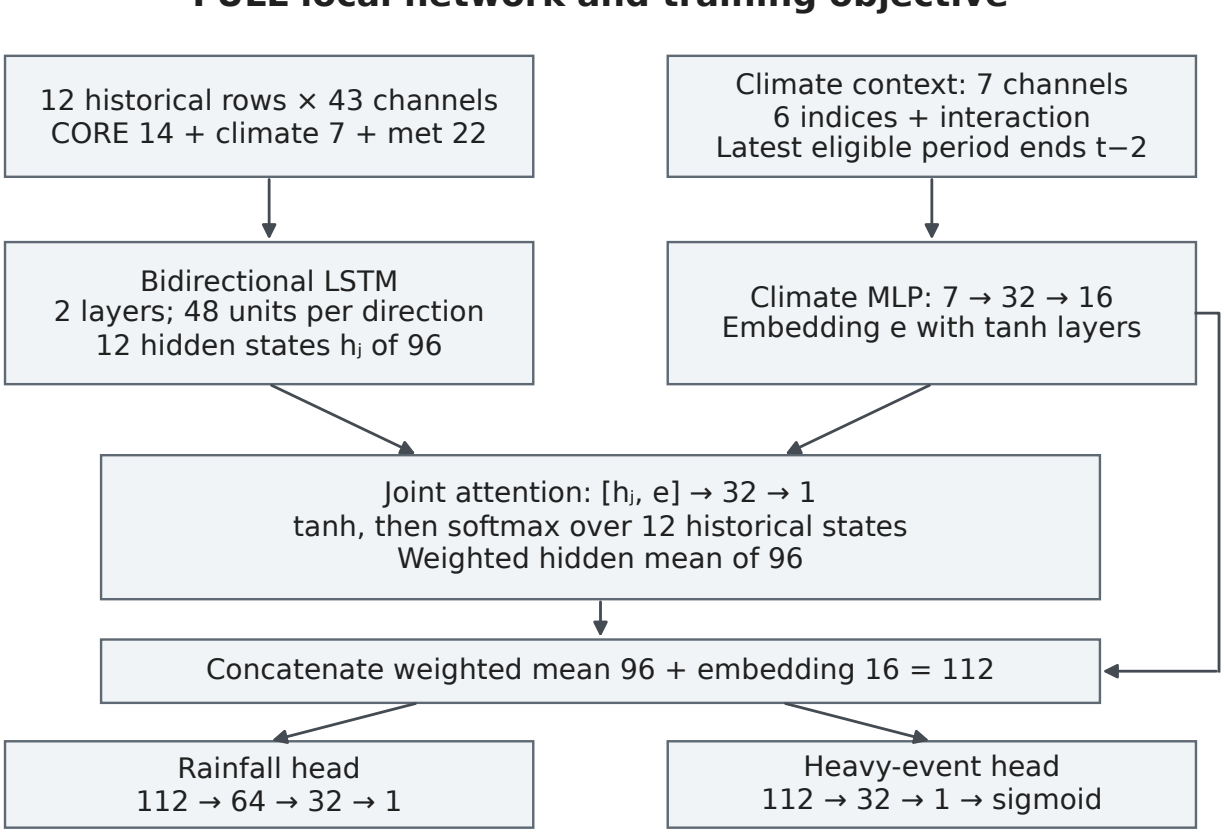

$$L_i = \mathrm{MSE}(y_i^z, \hat{y}_i^z) + 0.3\ \mathrm{BCE}(H_i, \hat{p}_i) + (\mu_i / 2)\ \|w_i - w^r\|_2^2$$

Shared TRAIN input scaling; local target scaling and q80.
CAAA batch proximity: $0.3\mu \le \mu_i \le \mu$.